



CrysDiS

Crystal Diffraction Simulator

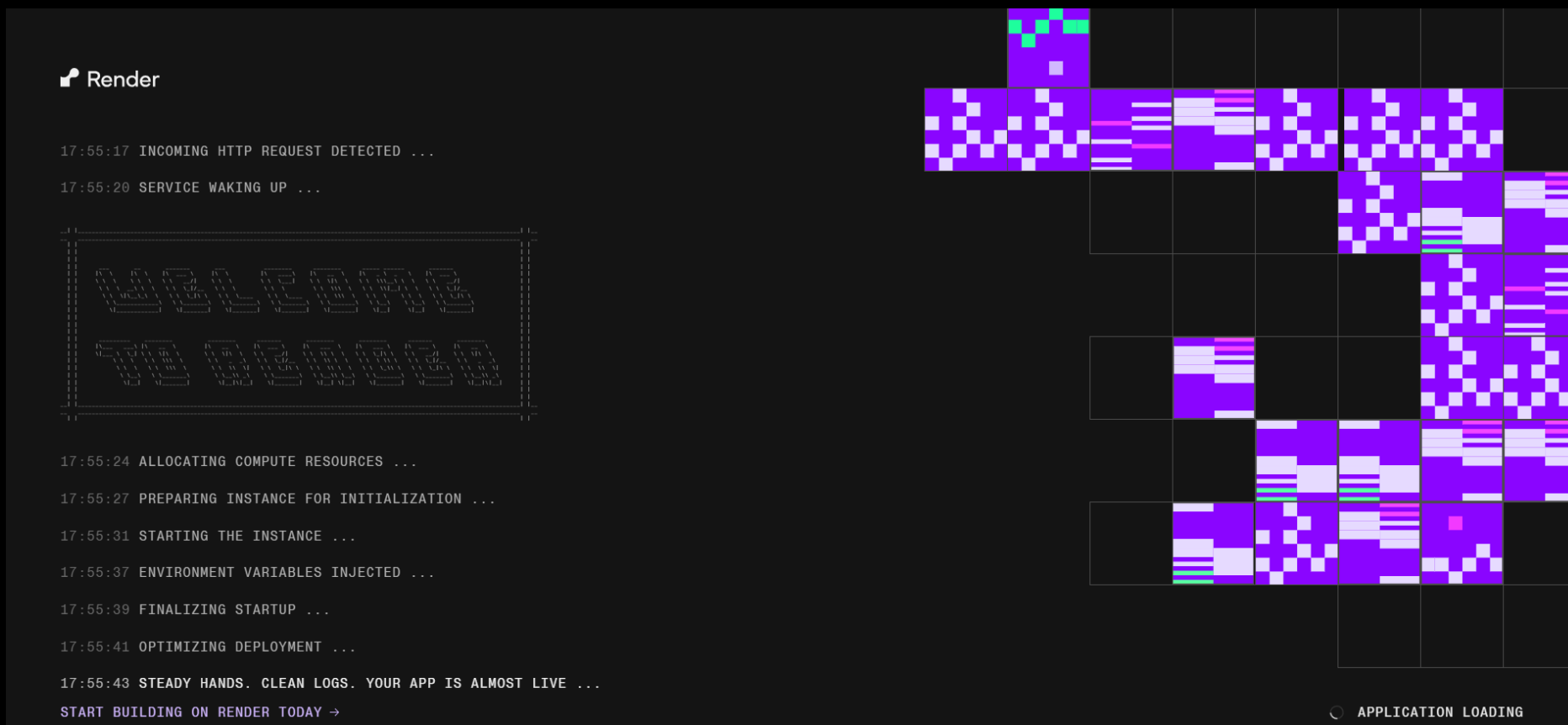
(created using ChatGPT Codex)



Start the program

Option 1: use the website version

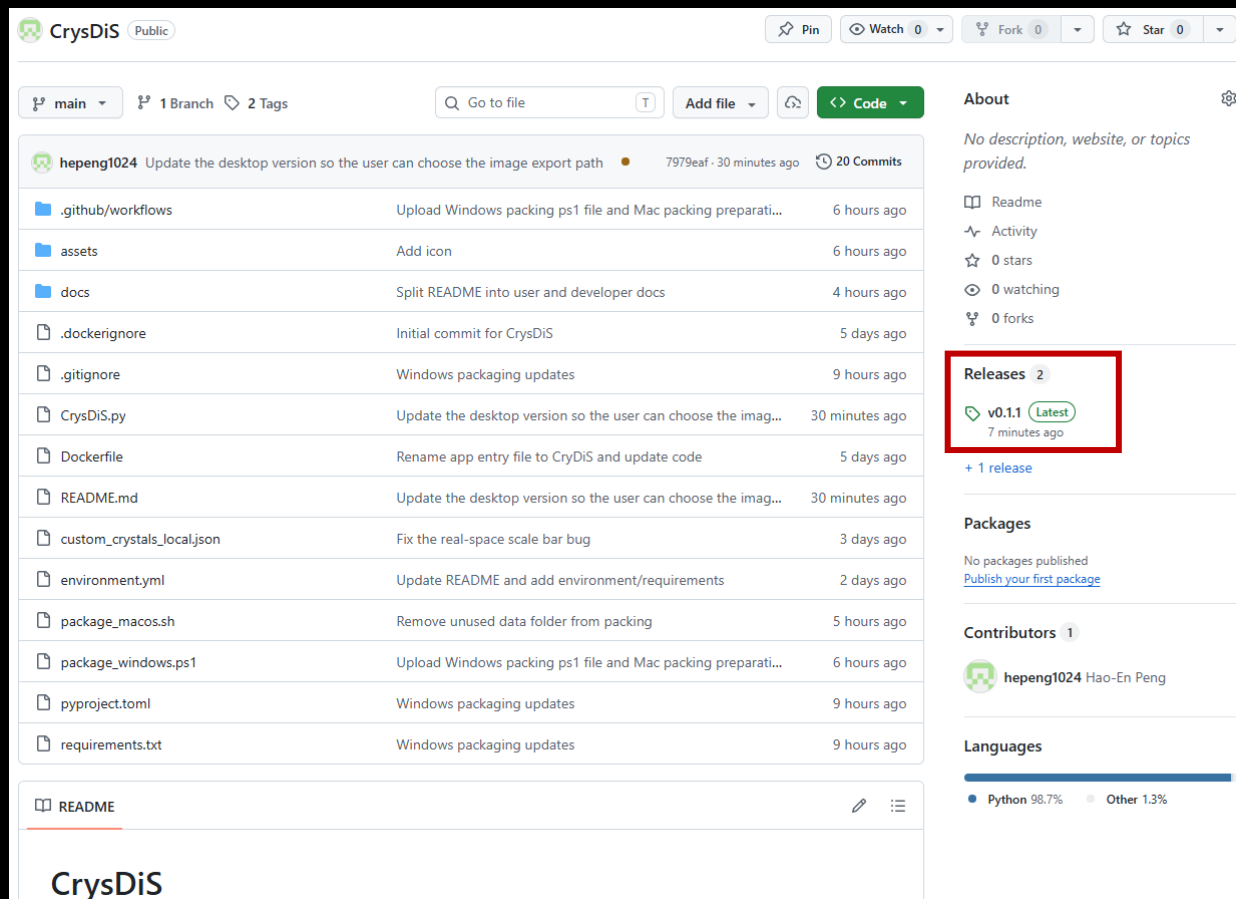
- Website: <https://crysdys.onrender.com/>
- Since this is a free URL, it may take 2~3 minutes to warm up
- The website version is not stable and may disconnect easily
- If you think this software is useful, running it locally is strongly recommended



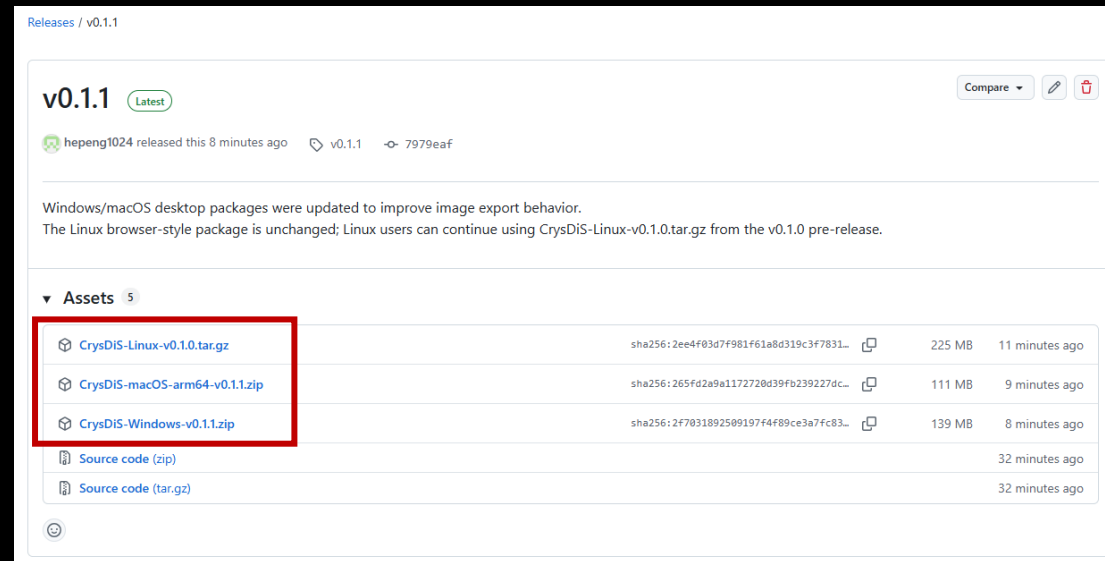
Start the program

Option 2: download the program and run it locally

- Go to <https://github.com/hepeng1024/CrysDiS> → Click “Releases” → Download the version that matches your OS



The screenshot shows the GitHub repository for CrysDiS. The 'Releases' section is highlighted with a red box, indicating the latest version is v0.1.1, released 7 minutes ago. The repository has 20 commits and 0 forks. The 'About' section states that no description, website, or topics are provided. The 'Packages' section shows no published packages. The 'Contributors' section lists hepeng1024 (Hao-En Peng). The 'Languages' section shows Python at 98.7% and Other at 1.3%.



The screenshot shows the release page for CrysDiS v0.1.1. The 'Assets' section is highlighted with a red box, showing three downloadable packages: CrysDiS-Linux-v0.1.0.tar.gz (225 MB, 11 minutes ago), CrysDiS-macos-arm64-v0.1.1.zip (111 MB, 9 minutes ago), and CrysDiS-Windows-v0.1.1.zip (139 MB, 8 minutes ago). The release notes mention that Windows/macOS desktop packages were updated to improve image export behavior, while the Linux browser-style package remains unchanged.

- Please refer to the README file if you need more information

Start the program

Option 3: Run the source code from GitHub

- First install Git and Anaconda/Miniconda. Then clone the repository:

```
git clone https://github.com/hepeng1024/CrysDiS.git
```

```
cd CrysDiS
```

- Create the conda environment:

```
conda env create -f environment.yml
```

```
conda activate crysdis
```

- Start CrysDiS:

```
python CrysDiS.py
```

- Then open this address in a browser:

```
http://127.0.0.1:8080
```

- If port 8080 is already being used, choose another port:

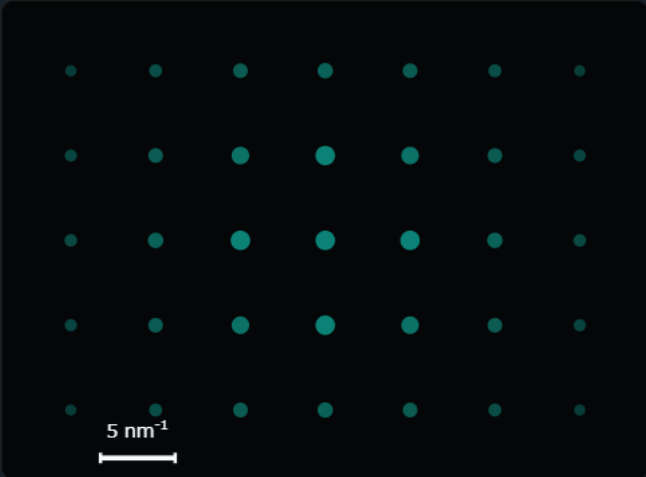
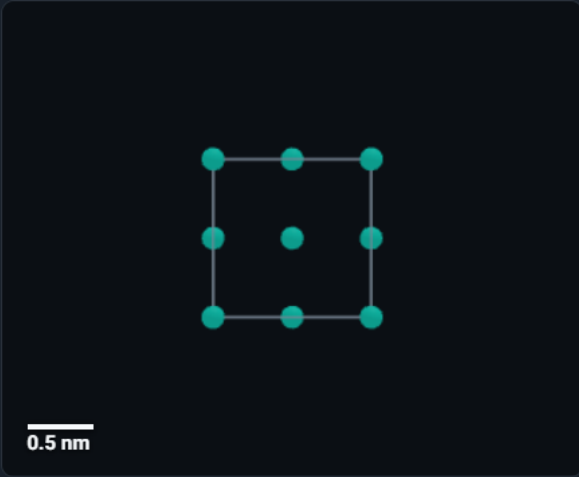
```
NICEGUI_PORT=8094 python CrysDiS.py
```

- Please refer to the README file at <https://github.com/hepeng1024/CrysDiS> for more details

Starting interface

CrysDiS [? INTRO](#) ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar [ADVANCED](#) [CRYSTAL LIST](#) [LOAD CIF](#) [CRYSTAL BUILDER](#) [ADD COMBO PANEL](#) [ADD PANEL](#)

1 Crystal **FCC** Zone axis **100** Plane Vector Rotation **APPLY** **SYNC**  

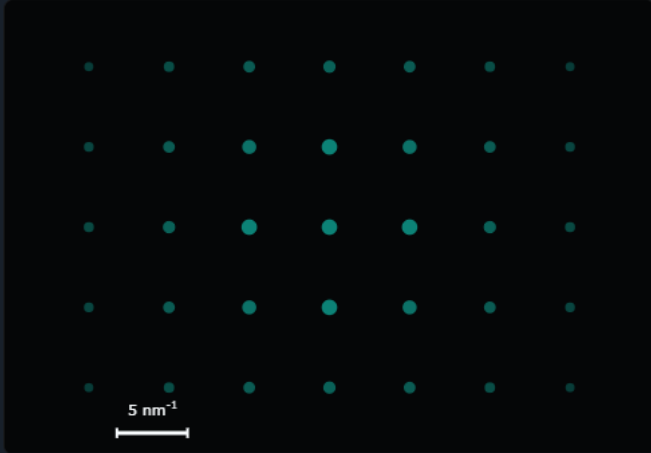
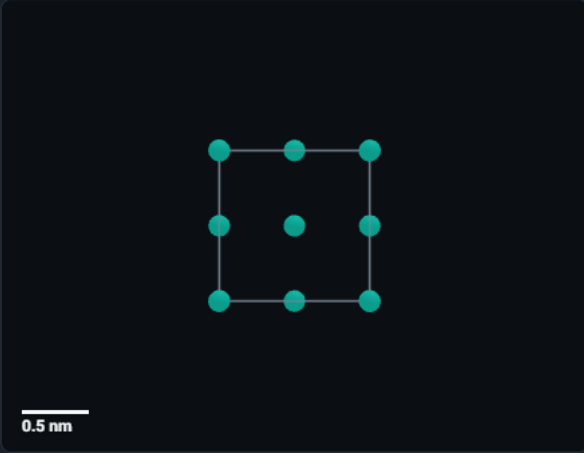


Features of the top toolbar

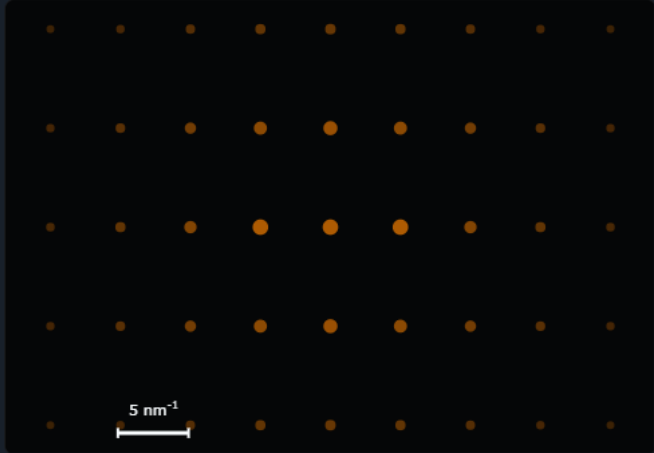
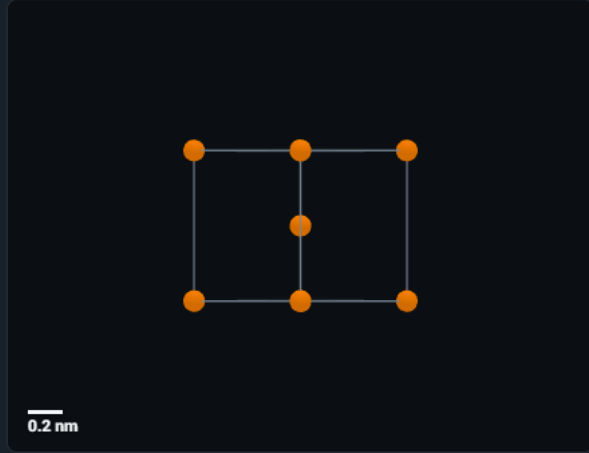
“ADD PANEL” allows you to have multiple structures at the same time

CrysDIS [?](#) [INTRO](#) ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar [ADVANCED](#) [CRYSTAL LIST](#) [LOAD CIF](#) [CRYSTAL BUILDER](#) [ADD COMBO PANEL](#) **ADD PANEL**

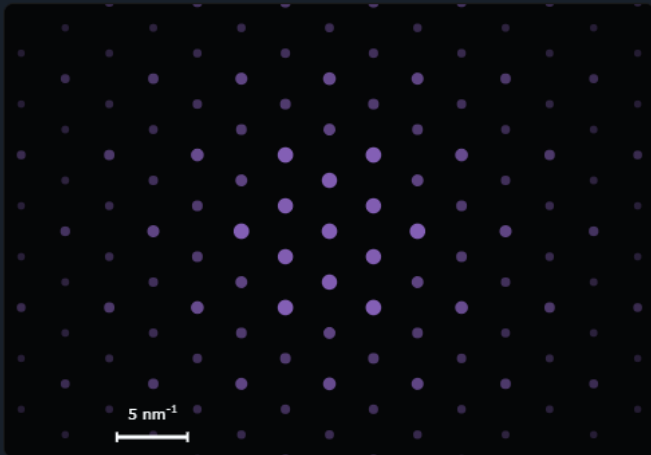
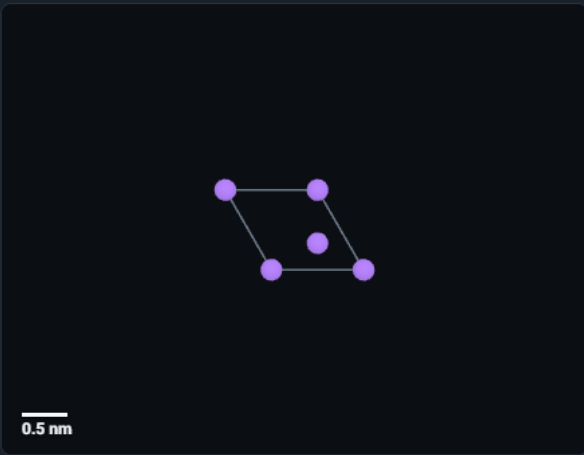
1 Crystal: FCC Zone axis: 100 Plane: Vector: Rotation: [APPLY](#) [SYNC](#)



2 Crystal: BCC Zone axis: 110 Plane: Vector: Rotation: [APPLY](#) [SYNC](#)



3 Crystal: HCP Zone axis: 0001 Plane: Vector: Rotation: [APPLY](#) [SYNC](#)

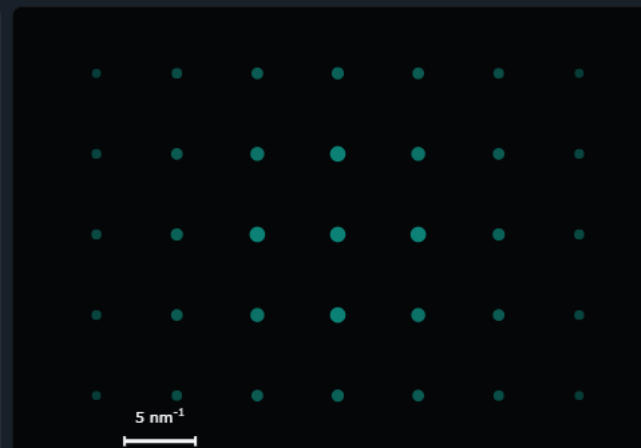
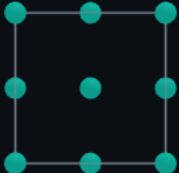


Features of the top toolbar

“ADD COMBO PANEL” allows you to overlap diffraction patterns from different panels and compare them

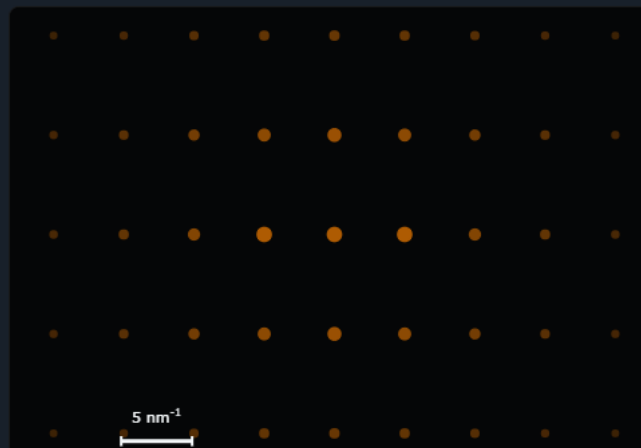
CrysDis [?](#) [INTRO](#) ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar [ADVANCED](#) [CRYSTAL LIST](#) [LOAD CIF](#) [CRYSTAL BUILDER](#) **ADD COMBO PANEL** [ADD PANEL](#)

1 Crystal: FCC Zone axis: 100 Plane: Vector: Rotation: [APPLY](#) [SYNC](#)



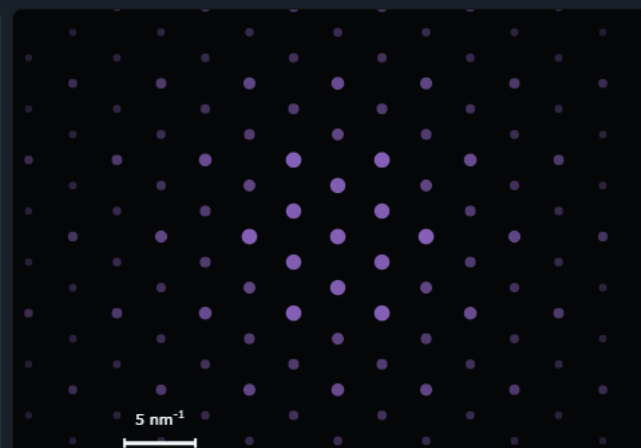
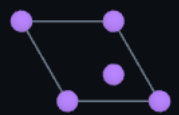
0.2 nm

2 Crystal: BCC Zone axis: 110 Plane: Vector: Rotation: [APPLY](#) [SYNC](#)



0.2 nm

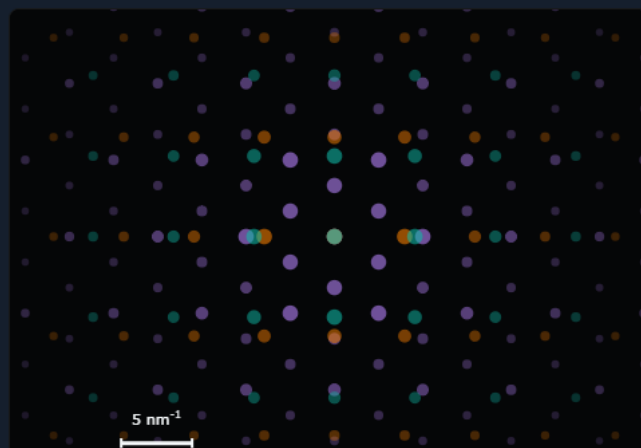
3 Crystal: HCP Zone axis: 0001 Plane: Vector: Rotation: [APPLY](#) [SYNC](#)



0.5 nm

C1 Current crystals: 1. FCC [ADD](#) [REMOVE](#) ☐ Bind crystal motion

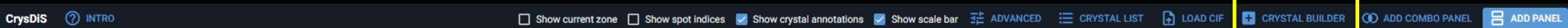
Panel	Crystal	Zone	
1	FCC	[100]	↑ ↓ ×
2	BCC	[110]	↑ ↓ ×
3	HCP	[0001]	↑ ↓ ×



5 nm⁻¹

Features of the top toolbar

“CRYSTAL BUILDER” lets you build your own crystal



Customized crystal builder

Name Customized crystal	Lattice cubic	Symmetry 1: P1	Save target Project local library	a 0.35 nm
b 0.35 nm	c 0.35 nm	alpha 90 deg	beta 90 deg	gamma 90 deg

APPLY LATTICE CONSTRAINTS	ADD ATOM SITE	PREVIEW SYMMETRY
---------------------------	---------------	------------------

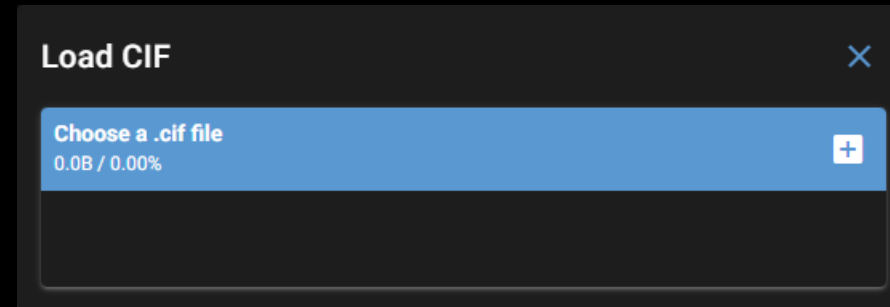
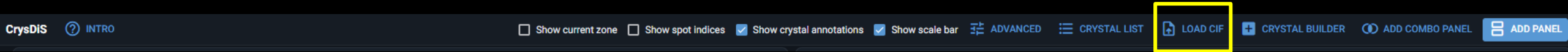
Atom sites

Atom Ni	x 0	y 0	z 0	Occ. 1	Color #10B7A5	Site Ni
------------	--------	--------	--------	-----------	------------------	------------

SAVE AS A NEW STRUCTURE

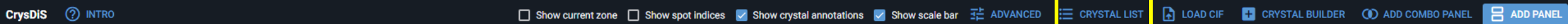
Features of the top toolbar

You can load your own .cif files via “LOAD CIF”



Features of the top toolbar

Manage the crystal structures via “CRYSTAL LIST”



Three built-in structures that cannot be edited

Crystal list			×
	FCC Built-in · cubic, Fm-3m, 1 site(s)		
	BCC Built-in · cubic, Im-3m, 1 site(s)		
	HCP Built-in · hexagonal, P1, 2 site(s)		
	(Al,Ti)Ni₃ Custom · cubic, P1, 5 site(s)		
	Al (mp-134) Custom · cubic, Fm-3m, 1 site(s)		
	AlNi₃ (mp-2593) Custom · cubic, Pm-3m, 2 site(s)		
	D024 Ni₃Ti Custom · hexagonal, P6 ₃ /mmc, 4 site(s)		
	L10 MnNi Custom · tetragonal, P4/mmm, 3 site(s)		
	L12 Ni₃Ti Custom · cubic, Pm-3m, 2 site(s)		
	Mg (mp-153) Custom · hexagonal, P6 ₃ /mmc, 2 site(s)		
	Mg₅Si₆ (mp-568306) Custom · monoclinic, C2/m, 6 site(s)		
	Mg₅Si₆_no_sym (mp-568306) Custom · triclinic, P1, 11 site(s)		
	Ni₃Si (mn-828)		

Features of the top toolbar

Advanced settings in “ADVANCED”

CrysDiS [INTRO](#) ☐ Show current zone ☐ Show spot indices ☒ Show crystal annotations ☒ Show scale bar **ADVANCED** [CRYSTAL LIST](#) [LOAD CIF](#) [CRYSTAL BUILDER](#) [ADD COMBO PANEL](#) [ADD PANEL](#)

Advanced settings

Simulation method
Pymatgen TEMCalculator

Camera length
200 mm

Voltage
500 kV

Thickness
10 nm

Max hkl
7

Spot intensity threshold
0.001

Compression factor
100
Contrast of diffraction spots intensity

Spot size scaling
0.3
How large bright spots are

☒ Always snap back view
Check to rotate the crystal back when moving to a new zone

☒ Use four-index basis for hexagonal systems

☐ Auto sync crystal/diffraction

PALETTE FOR VECTORS/PLANES

Log
"1 123" is not a valid index for FCC
Panel 1: snapped back to [100]
"1- 123" is not a valid index for FCC
Panel 1: snapped back to [100]
Panel 1: snapped back to [100]

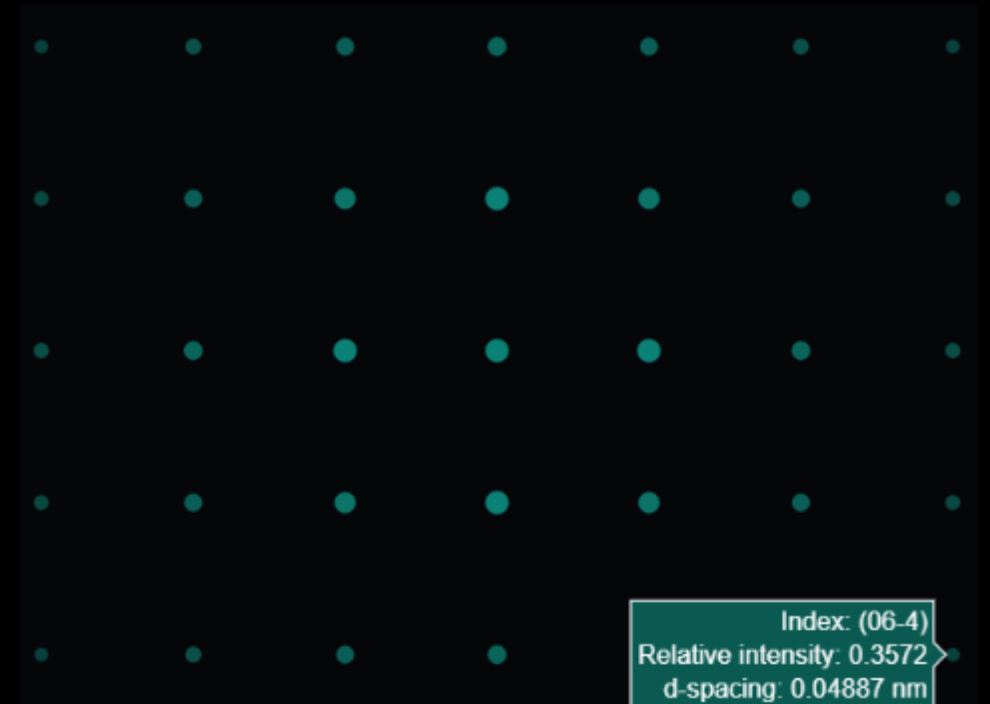
More about advanced settings

- Compression factor

Compression factor = 10



Compression factor = 150



Dim spots are more visible

More about advanced settings

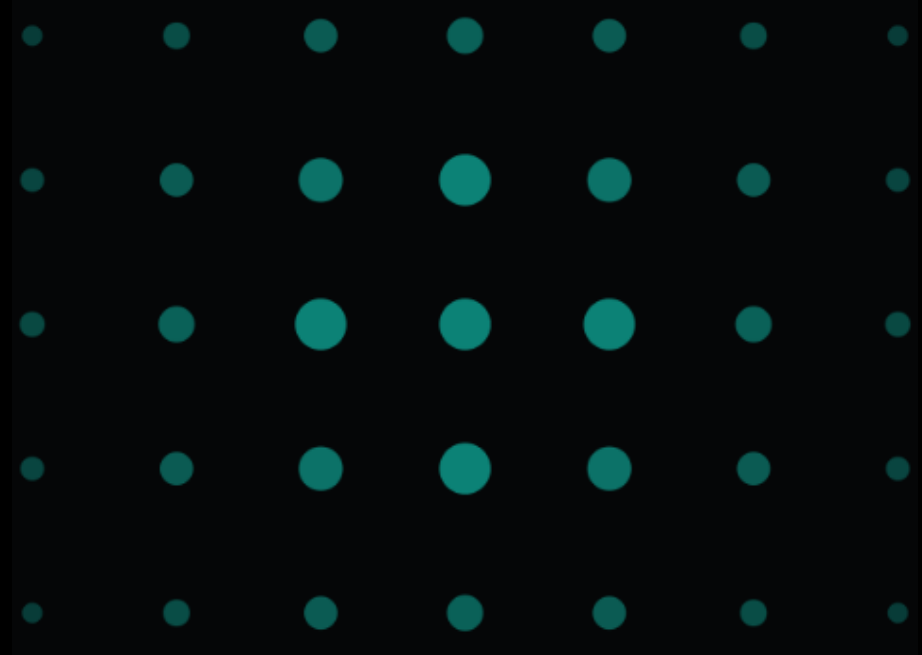
- Spot size scaling effect

Scaling factor = 0



All spots are the same size

Scaling factor = 1



Spot size scales significantly with intensity

More about advanced settings

- Always snap back view



In-plane rotation by 45 deg



Go to the 110 zone

“Always snap back view” is checked

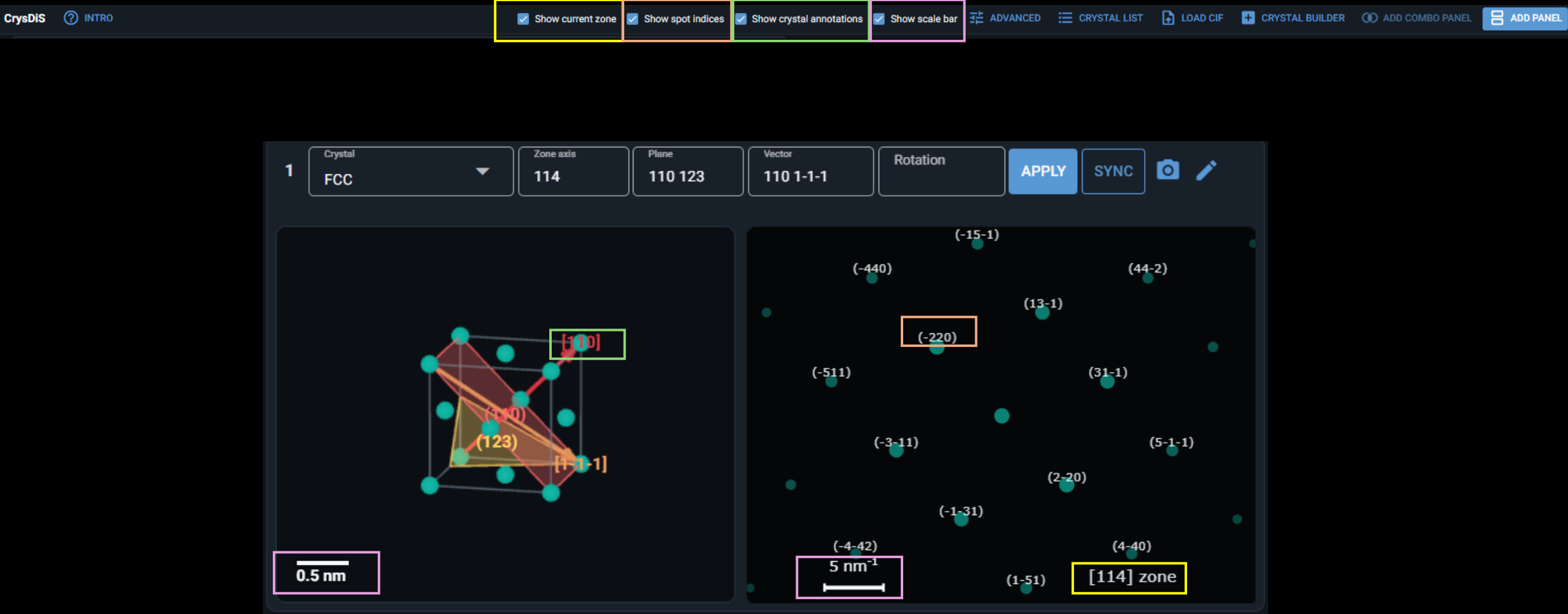


“Always snap back view” is unchecked



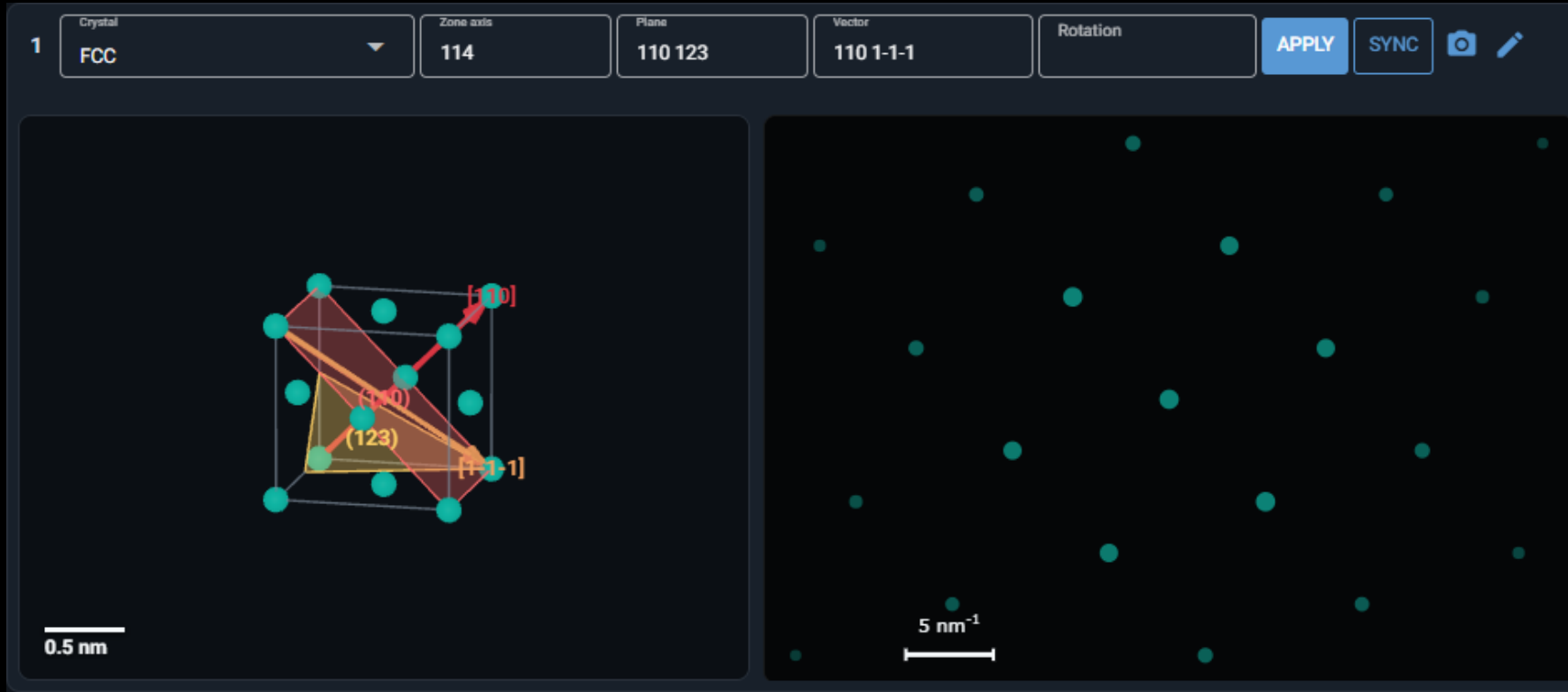
Features of the top toolbar

Some annotation toggles



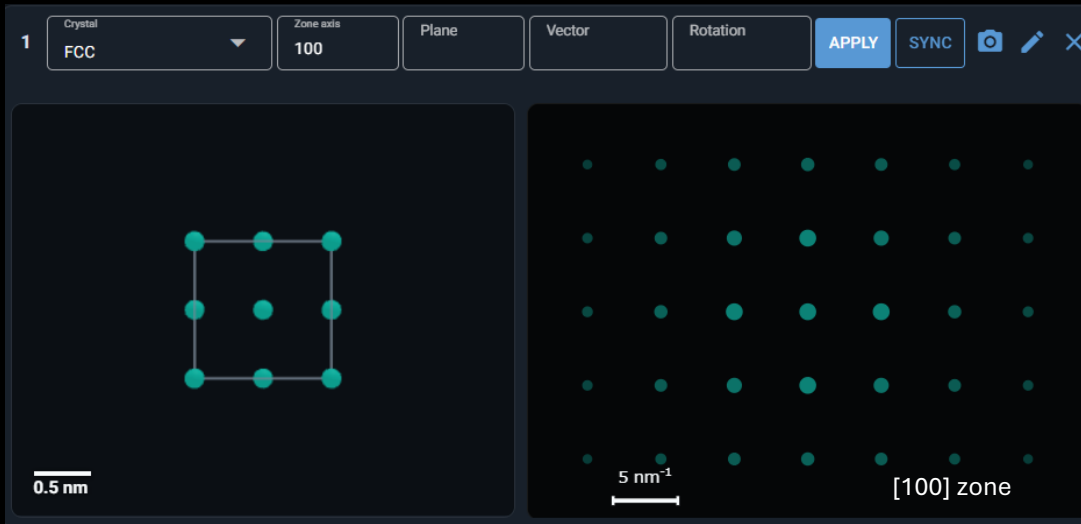
Features of the panel toolbar

- Choose any crystals in the list or upload/build a new one
- You can go to any zone axis, create vectors/planes
- Reciprocal lattice vectors can be created using *hkl (ex. *110, *0-10)
- For hexagonal systems, 4-index basis inputs are allowed (ex. 2-1-10, *0001)

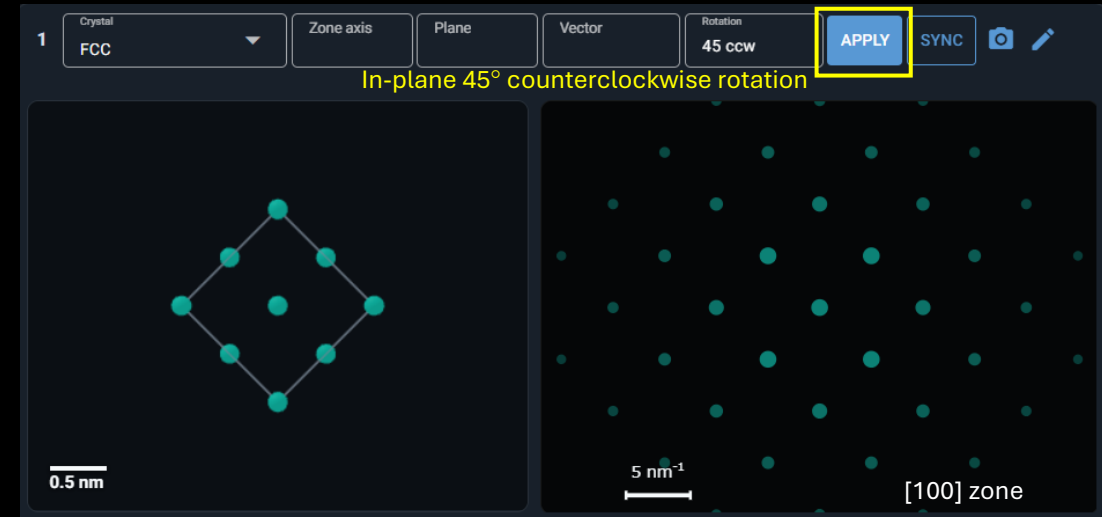


Features of the panel toolbar

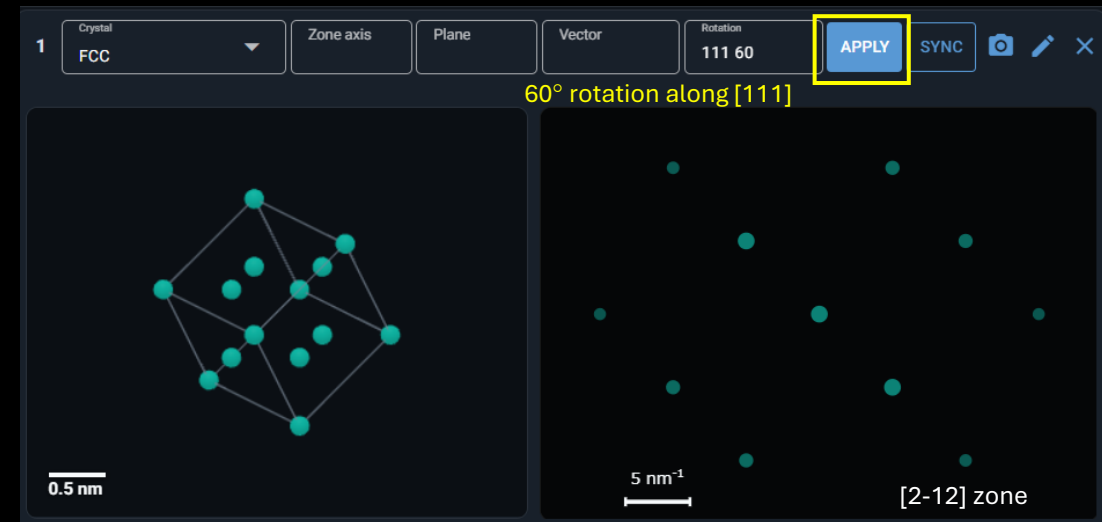
- You can also rotate the crystal along a certain axis
- Right-hand rule is applied



In-plane rotation: use “deg cw” or “deg ccw”

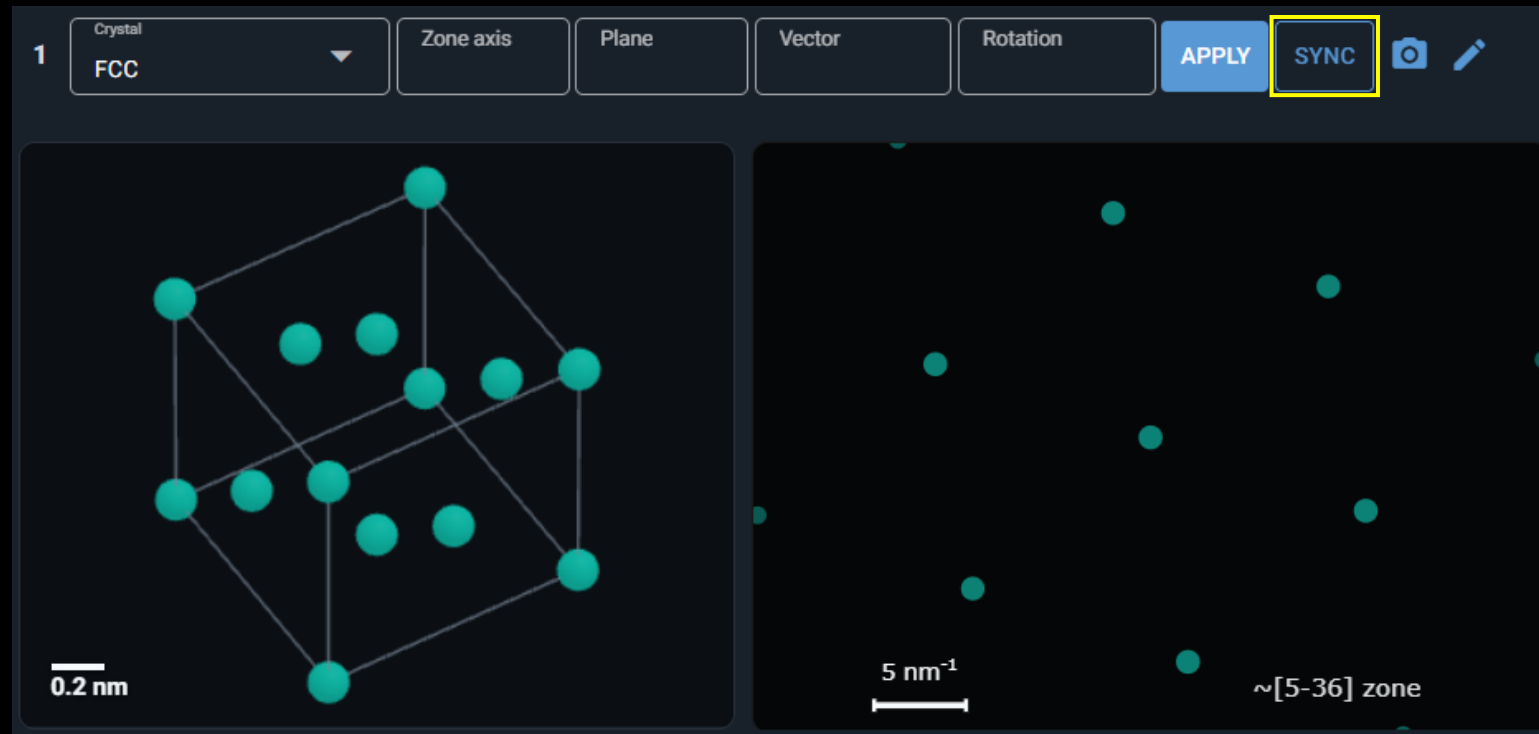


Rotation along a certain axis: use “vector deg”



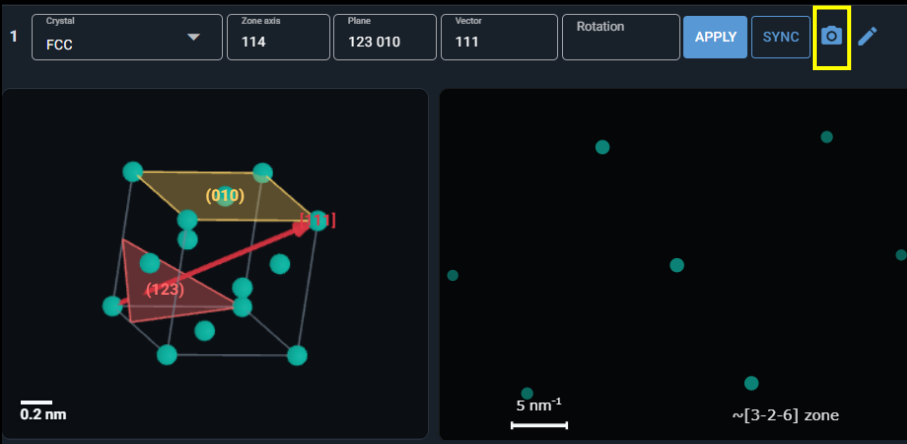
Features of the panel toolbar

- Synchronize the crystal and diffraction pattern using “SYNC”
- It will detect the closest zone axis
- You can also turn on the “Auto sync crystal/diffraction” in the advanced settings



Features of the panel toolbar

- Download the crystal/diffraction pattern at the current viewing angle



Download panel 1

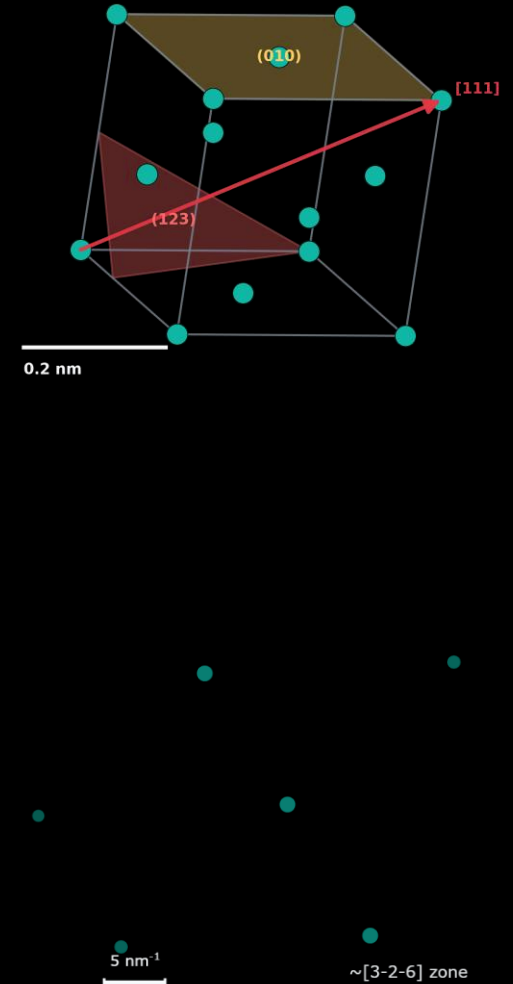
☒ Real-space crystal

☒ Diffraction pattern

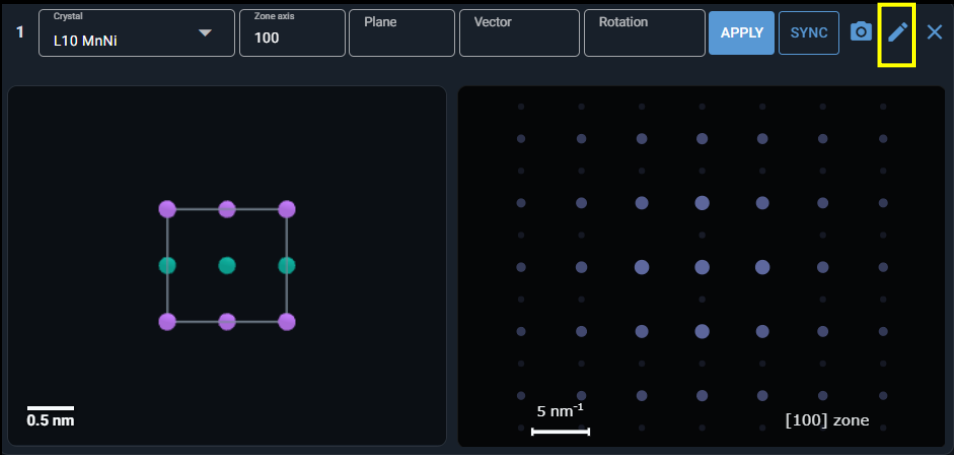
☒ Transparent background

Quality
300 dpi

CANCEL [DOWNLOAD]



Features of the panel toolbar



- Edit the current structure

Customized crystal builder

Name

L10 MnNi

Lattice

tetragonal

Symmetry

123: P4/mmm

Save target

Project local library

a

0.374

nm

b

0.374

nm

c

0.352

nm

alpha

90

deg

beta

90

deg

gamma

90.0

deg

⚙

APPLY LATTICE CONSTRAINTS

+

ADD ATOM SITE

⚙

PREVIEW SYMMETRY

Atom sites

Atom Mn	x 0	y 0	z 0	Occ. 1	Color #C77DFF		Site Mn1	
Atom Mn	x 0.5	y 0.5	z 0	Occ. 1	Color #C77DFF		Site Mn2	
Atom Ni	x 0.5	y 0	z 0.5	Occ. 1	Color #10B7A5		Site Ni1	

💾

SAVE EDITED STRUCTURE

💾

SAVE AS A NEW STRUCTURE

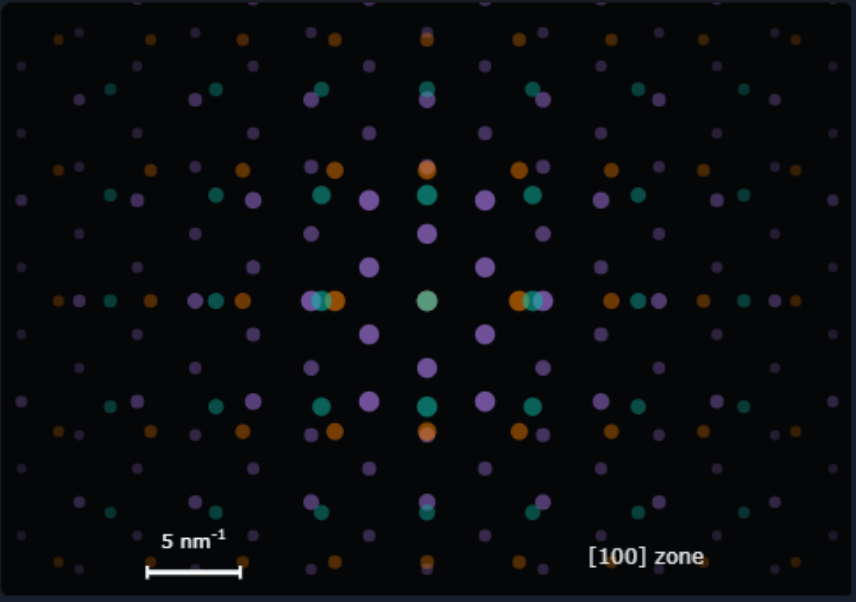
Features of the combo panel

Select crystals present on the page

Add or remove it

C1 Current crystals 1. FCC ADD REMOVE ☐ Bind crystal motion  ✕

Panel	Crystal	Zone	
1	FCC	[100]	↑ ↓ ✕
2	BCC	[110]	↑ ↓ ✕
3	HCP	[0001]	↑ ↓ ✕



5 nm⁻¹ [100] zone

The interface shows a list of crystals and their corresponding diffraction patterns. The current crystal is FCC, and the current zone is [100]. The diffraction pattern is a 2D grid of colored dots (purple, green, orange) on a dark background. A scale bar indicates 5 nm⁻¹ and the zone is labeled [100] zone.

Ex. FCC diffraction pattern will be shown on top

The current zone shows the zone axis of the crystal on the top-most layer

The order of the crystal in the list determines the order of the diffraction patterns

Features of the combo panel

Check “Bind crystal motion” if you want the crystals to synchronize their motions

C1

Current crystals
1. FCC

ADDREMOVE☒ Bind crystal motion📷✕

Panel	Crystal	Zone	
1	FCC	[100]	↑ ↓ ✕
2	BCC	[110]	↑ ↓ ✕

5 nm⁻¹

[100] zone

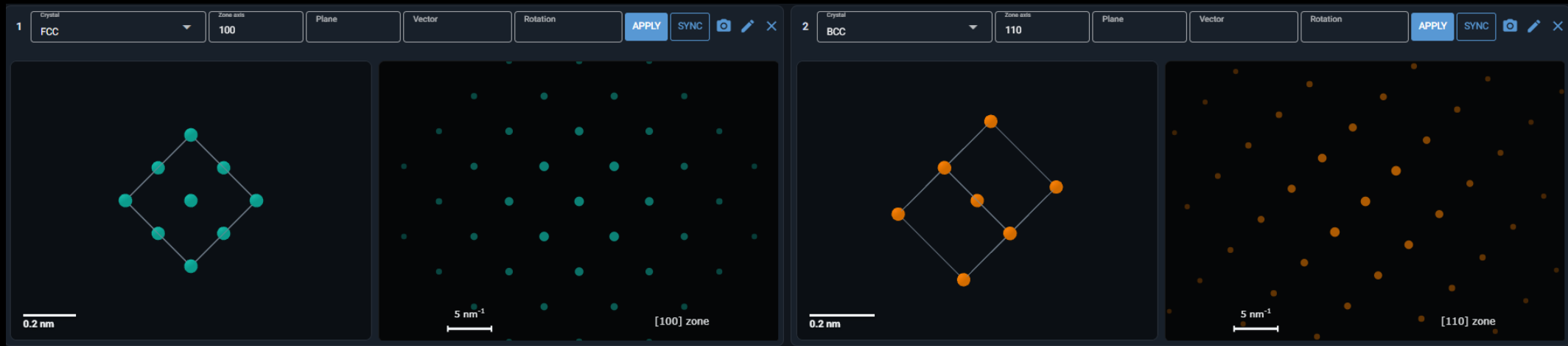
Features of the combo panel

- Bind crystal motion



Ex. if you rotate the crystal in panel 1 by 45° ccw

the crystal in panel 2 will do the same motion



In this case, it means crystals 1 & 2 have the orientation relationship: $[100]_{\text{crystal1}} // [110]_{\text{crystal2}}$

Comparison of two simulation methods

Simulation method	Ewald Sphere Kinematic	Pymatgen TEMCalculator
Main purpose	More TEM-like kinematic spot visibility	Fast crystallographic zone-axis pattern
Uses voltage?	Yes, through electron wavelength and Ewald sphere radius	Only for wavelength calculation
Uses thickness?	Yes	No
Includes relrod / excitation error?	Yes	No
Includes Ewald sphere curvature?	Yes	No
Good for	Showing why spots appear/disappear with tilt/thickness	Clean ideal zone-axis indexing/intensity